Trajectory Systems: Circular Verification Collapses Canonical Trust

Extracted from a machine-checked development

Abstract

A trajectory system is a trust system in which claims are agent actions and checking steps are causal dependencies among those actions. We introduce a five-level trajectory chain and two concrete systems sharing a common dependency topology: a *circular* trajectory, modelling circular verification, and an *honest* trajectory with sound grades. In the circular system, canonical trust τ evaluates to bottom at every event. In the honest system, τ attains the grade independent along sound paths and is limited by the weakest step grade along weaker ones. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

Trust systems grade claims by the strength of the checks that support them. Canonical trust τ is the greatest sound assignment of grades: at each claim, the best (join) over checking steps of the meet of the step's own grade with the canonical trusts of its premises. When the premise relation is well founded, τ is uniquely determined by well-founded recursion.

Trajectory systems specialise this picture to agent workflows. Claims are events (actions of an agent); checking steps record causal dependence of one event on others; grades record how strongly each dependence was checked. The motivating phenomenon is *circular verification*: a chain of events in which each step claims independence of prior checks, yet the root is unsupported and a terminal "verify" step is itself unsupported. The widest-bottleneck semantics of τ then forces every event to bottom, regardless of intermediate grades that look strong in isolation.

We work throughout with finite chains of check strengths and finite systems of events. No special order theory beyond meets and joins in a chain is required. Proofs are indicated by the phrase "by lean," meaning they have been checked by a proof assistant rather than written out in prose.

What this note claims. In ordinary mathematical language, the claim is: under the circular grading of Definition 5.1, canonical trust is identically bottom; under the honest grading of Definition 6.1 on the same dependency topology, canonical trust recovers the independent grade along the sound spine and is limited by the shallow grade at the weaker branch. Circular verification is therefore invisible to node-local grade inspection alone: it is diagnosed only by computing τ globally.

2 Recollections: chains and trust systems

We recall only what is needed; full development appears in the companion note on trust systems.

Definition 2.1 (Chain)**.** A **chain** T is a linearly ordered type equipped with a bottom element $\bot$, such that $\leq$ is reflexive, transitive, antisymmetric, and total, and $\bot \leq a$ for every $a \in T$. Meets and joins exist and are computable:

$$\operatorname{cmin}(a,b) \;=\; \begin{cases} a & \text{if } a \leq b, \\ b & \text{otherwise,} \end{cases} \qquad \operatorname{cmax}(a,b) \;=\; \begin{cases} b & \text{if } a \leq b, \\ a & \text{otherwise.} \end{cases} \tag{1}$$

Definition 2.2 (Trust system)**.** A **trust system** over claims C graded in a chain T consists of

1. a type Step of checking steps,
2. a conclusion map $\mathrm{concl} : \mathrm{Step} \to C$,
3. a finite list of premises $\mathrm{prems} : \mathrm{Step} \to \mathrm{List}(C)$,
4. a grade map $\mathrm{grade} : \mathrm{Step} \to T$,
5. for each claim x, a finite list $\mathrm{stepsInto}(x)$ of steps concluding at x, coherent with concl.

The **premise relation** $y \prec x$ holds when y is a premise of some step into x.

Definition 2.3 (Step grade and canonical trust)**.** Given an assignment $t : C \to T$, the **step grade** of a step f is

$$\mathrm{stepGrade}(t,f) \;=\; \operatorname{cmin}\big(\mathrm{grade}(f),\ \operatorname{cmin}_{y \in \mathrm{prems}(f)} t(y)\big) \tag{2}$$

(the meet of the step's own grade with the assigned levels of all premises; empty meets yield the step grade alone).

When the premise relation is well founded, **canonical trust** $\tau : C \to T$ is defined by well-founded recursion:

$$\tau(x) \;=\; \operatorname{cmax}\Big(\bot,\ \operatorname{cmax}_{f \in \mathrm{stepsInto}(x)} \mathrm{stepGrade}(\tau, f)\Big). \tag{3}$$

Equivalently, $\tau(x)$ is the join over steps into x of their step grades under τ itself, defaulting to $\bot$ when there are no steps.

Remark 2.4. Canonical trust is the greatest sound assignment: every sound grading is bounded by τ, and τ itself is sound. We use only the recursive character of τ below.

3 The trajectory chain

Definition 3.1 (Trajectory levels)**.** The **trajectory chain** T_{traj} is the five-element chain

$$\mathsf{unsupported} \;<\; \mathsf{shallow} \;<\; \mathsf{independent} \;<\; \mathsf{crossVerified} \;<\; \mathsf{mechChecked}, \tag{4}$$

with bottom $\bot = \mathsf{unsupported}$. The order is the unique linear order extending the displayed chain of covering relations (and equality).

Proposition 3.2 (Chain structure)**.** T_{traj} *carries the structure of a chain in the sense of Definition 2.1.*

Proof. By lean. □

Remark 3.3. Informally: unsupported means no usable check; shallow a weak or one-sided check; independent a check that does not depend on the claim under review; crossVerified independent confirmation from a second source; mechChecked machine verification.

4 Events and shared topology

Definition 4.1 (Events)**.** Let $E = \{e_0, e_1, e_2, e_3, e_4\}$ be a five-element type of **events**.

Both systems below use the same dependency topology: steps are identified with events (one step concluding at each event), and premises form a directed acyclic graph

$$e_0 \longrightarrow e_1 \longrightarrow e_2 \longrightarrow e_3, \qquad e_2 \longrightarrow e_4. \tag{5}$$

Explicitly:

Event	Premises	Informal role
e_0	$\emptyset$	root / ground event
e_1	$\{e_0\}$	first dependent step
e_2	$\{e_1\}$	intermediate step
e_3	$\{e_2\}$	claim branch
e_4	$\{e_2\}$	verify branch

Proposition 4.2 (Well-foundedness of the topology)**.** *The premise relation of this topology is well founded.*

Proof. By lean. □

5 The circular trajectory

Definition 5.1 (Circular trajectory)**.** The **circular trajectory** is the trust system $\mathcal{S}_{\text{circ}}$ on claims E graded in T_{traj} with the topology of the previous section and grades

Event	Grade
e_0	unsupported
e_1	independent
e_2	independent
e_3	shallow
e_4	unsupported

Remark 5.2 (Why "circular")**.** The topology is acyclic; circularity lives in the grades. The root e_0 is unsupported, yet e_1 and e_2 are graded as independent checks, and the terminal verify step e_4 is itself unsupported. Locally, intermediate steps look strong; globally, every path either begins at an unsupported root or ends at an unsupported verify. The computation of τ makes this precise.

Lemma 5.3 (Leaf computation)**.** *In* $\mathcal{S}_{\text{circ}}$*,*

$$\tau(e_0) \;=\; \text{unsupported}. \tag{6}$$

Proof. By lean. □

Lemma 5.4. *In* $\mathcal{S}_{\mathrm{circ}}$*,*

$$\tau(e_1) \;=\; \mathsf{unsupported}. \tag{7}$$

Proof. By lean. □

Lemma 5.5. *In* $\mathcal{S}_{\mathrm{circ}}$*,*

$$\tau(e_2) \;=\; \mathsf{unsupported}. \tag{8}$$

Proof. By lean. □

Lemma 5.6. *In* $\mathcal{S}_{\mathrm{circ}}$*,*

$$\tau(e_3) \;=\; \mathsf{unsupported}. \tag{9}$$

Proof. By lean. □

Lemma 5.7. *In* $\mathcal{S}_{\mathrm{circ}}$*,*

$$\tau(e_4) \;=\; \mathsf{unsupported}. \tag{10}$$

Proof. By lean. □

Theorem 5.8 (Circular collapse)**.** *In the circular trajectory, canonical trust is bottom at every event:*

$$\forall\, e \in E, \qquad \tau_{\mathcal{S}_{\mathrm{circ}}}(e) \;=\; \mathsf{unsupported}. \tag{11}$$

Proof. By lean. □

Remark 5.9 (Idea of the argument)**.** Proceed leaf-to-root along the premise order. At e_0 the unique step has empty premises and grade unsupported, so $\tau(e_0) = \mathsf{unsupported}$. At e_1 the step grade is $\mathrm{cmin}(\tau(e_0), \mathsf{independent}) = \mathsf{unsupported}$, and similarly for e_2. At e_3 one meets $\tau(e_2)$ with shallow; at e_4 one meets $\tau(e_2)$ with unsupported. In every case the join with $\bot$ remains unsupported. Intermediate grades of independent never appear in τ because they are bottlenecked by unsupported ancestors (or, for e_4, by an unsupported self-grade).

6 The honest trajectory

Definition 6.1 (Honest trajectory)**.** The **honest trajectory** is the trust system $\mathcal{S}_{\mathrm{hon}}$ on the same claims E, the same topology, and grades

Event	Grade
e_0	independent
e_1	independent
e_2	independent
e_3	shallow
e_4	independent

Proposition 6.2 (Well-foundedness)**.** *The premise relation of* $\mathcal{S}_{\mathrm{hon}}$ *is well founded.*

Proof. By lean. □

Theorem 6.3 (Honest root)**.** *In* $\mathcal{S}_{\mathrm{hon}}$*,*

$$\tau(e_0) \;=\; \mathsf{independent}. \tag{12}$$

Proof. By lean. □

Theorem 6.4 (Honest spine)**.** *In* $\mathcal{S}_{\text{hon}}$*,*

$$\tau(e_1) \;=\; \mathsf{independent}. \tag{13}$$

Proof. By lean. □

Theorem 6.5. *In* $\mathcal{S}_{\text{hon}}$*,*

$$\tau(e_2) \;=\; \mathsf{independent}. \tag{14}$$

Proof. By lean. □

Theorem 6.6 (Weak claim branch)**.** *In* $\mathcal{S}_{\text{hon}}$*,*

$$\tau(e_3) \;=\; \mathsf{shallow}. \tag{15}$$

Proof. By lean. □

Theorem 6.7 (Honest verify branch)**.** *In* $\mathcal{S}_{\text{hon}}$*,*

$$\tau(e_4) \;=\; \mathsf{independent}. \tag{16}$$

Proof. By lean. □

Remark 6.8 (Contrast with the circular case)**.** The topology is identical; only the grades differ. With an independent root and independent steps along the spine $e_0 \to e_1 \to e_2 \to e_4$, canonical trust preserves independent. The claim branch e_3 is limited by its own shallow grade: weakest-link semantics yields $\mathrm{cmin}(\tau(e_2), \mathsf{shallow}) = \mathsf{shallow}$. Thus τ distinguishes sound verification structure from circular structure on the nose.

7 Discussion

What has been established. Trajectory systems are trust systems specialised to agent-action claims. On a fixed five-event topology, circular grades force $\tau \equiv \mathsf{unsupported}$, while honest grades recover independent along the sound path and shallow on the weak branch. The diagnosis of circular verification is therefore a theorem about τ, not a side condition on individual step grades.

Role of well-foundedness. Both systems have well-founded premise relations (the dependency graph is a finite DAG). Well-foundedness is load-bearing for the existence of τ as a recursive definition; circular *citation* in the sense of a cyclic premise relation is a different pathology (treated in the companion note on trust systems). Here the circularity is graded rather than structural.

Scope. The development treats a single finite event set and a single five-level chain. It does not address infinite trajectories, continuous time, or composition of trajectory systems with other trust systems; the latter is taken up in a companion note on trajectory–trust morphisms.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant; we have not reproduced the formal proof scripts here. A mathematician reading this note need only verify that the English statements match the intended mathematics; the formal system is responsible for the derivations.

Acknowledgements

The mathematical content specialises the theory of trust systems (canonical trust as greatest sound assignment, well-founded premise recursion) to trajectory workflows. The circular example is motivated by observed agent behaviour under circular verification. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

[1] Trust systems: the certificate lemma and transport of trust. Companion formal development.